Special Topics for EEM Assignment - 2

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Q1)

$$\theta = 1,22 \frac{500 \times 10^{-9}\ nm}{8 \times 10^{-2}\ nm}$$

$$= 7,625 \times 10^{-6}\ rad$$

Q2)

$y = m\lambda L / d$

$d = 100m$, $L = 5m$ $\Rightarrow$ $y = 0 \times 460nm \times 5m / 10m \Rightarrow y = 0m$ for $m=0$

$y = 1 \times 460 \times 5m / 10m \Rightarrow y = 0,023m$ for $m=1$

$y = 2 \times 460 \times 5m / 10m \Rightarrow y = 0,046m$ for $m=2$

The first three bright spot on the screen are = $y=0$, $y=0,23$, $y=0,046$

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Q4

$$r_{TE} = \frac{n_1\cos\theta_i - n_2\cos\theta_t}{n_1\cos\theta_i + n_2\cos\theta_t} \quad , \quad t_{TE} = \frac{2n_1\cos\theta_i}{n_1\cos\theta_i + n_2\cos\theta_t}$$

When we use Snell's Law $\Rightarrow n_1\sin\theta_i = n_2\sin\theta_t \Rightarrow \sin\theta_t = \frac{n_1}{n_2}\sin\theta_i$

$$r_{TE} = \frac{n_1\cos\theta_i - n_2\sqrt{1-\left(\frac{n_1}{n_2}\sin\theta_i\right)^2}}{n_1\cos\theta_i + n_2\sqrt{1-\left(\frac{n_1}{n_2}\sin\theta_i\right)^2}}$$

$$t_{TE} = \frac{2n_1\cos\theta_i}{n_1\cos\theta_i + n_2\sqrt{1-\left(\frac{n_1}{n_2}\sin\theta_i\right)^2}}$$

$$T = \frac{n_2\cos\theta_t}{n_1\cos\theta_i}|t_{TE}|^2 = 0{,}95 \quad , \quad \theta_i = \arcsin\left(\frac{n_2}{n_1}\sqrt{\frac{0{,}95}{2}}\right) \cong 41{,}6$$

Q5

$I = I_0\cos^2\theta$

$\Delta\theta = 30^\circ - 0^\circ = 30^\circ \Rightarrow I_1 = 800\times\cos 30 = 600\ \frac{w}{m^2}$

$I_2 = I_1\cos^2\theta \Rightarrow 600\times\cos^2 30 = 450\ \frac{w}{m^2}$

$\theta = 75^\circ - 30^\circ = 45^\circ$

$I_3 = 450\times\cos^2 45 = 225\ \frac{w}{m^2}$

Q7

$$F_r = \frac{2I}{c}A = \frac{2(1350)}{3(10^8)}A = 9\times10^{-6}A \qquad \frac{F}{mg} = \frac{9\times10^{-6}}{9{,}8} = 9{,}18\times10^{-7}A$$